Marketplace Performance Analysis

Business Data Analyst Portfolio Project

Dataset: 60 Sellers | 600 Orders | Jan 2023 - Dec 2025

Tools: PostgreSQL | Excel | Power BI | PowerPoint

Analyst: Tushar Kumar

PROJECT OVERVIEW

BUSINESS CONTEXT

Back Market is a two-sided marketplace for refurbished electronics, connecting individual buyers with professional refurbishers across Europe and North America.

This project simulates the analytical work of a Business Data Analyst inside Back Market - using real marketplace KPIs and business logic to answer practical commercial questions.

5 BUSINESS QUESTIONS ANSWERED

Q1 Which product categories generate the most revenue?

Q2 Which countries drive the most orders and revenue?

Q3 Which sellers have the highest return rates?

Q4 Which categories have the most returned orders?

Q5 Which sellers had the highest completed order volume?

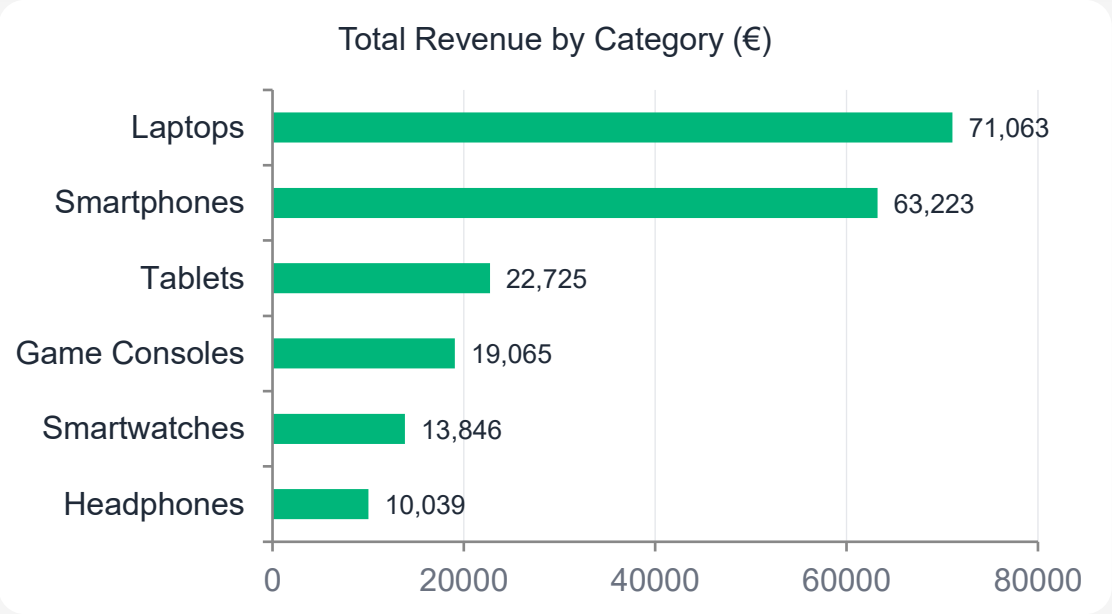
Q1 - REVENUE BY PRODUCT CATEGORY

PostgreSQL + Excel

€199.96K
Total Revenue

493
Completed Orders

€323.50
Avg Order Value



KEY INSIGHTS

- Laptops lead revenue at €71K despite fewer orders (97) than Smartphones (152)
- Laptops avg order value (€676) is nearly 2× Smartphones (€355)
- Top 2 categories (Laptops + Smartphones) = 67% of total revenue
- Headphones drive lowest revenue at €10K - lowest priority category

▶ Prioritise Laptop category investment - highest revenue per transaction

Q2 - GEOGRAPHIC PERFORMANCE

PostgreSQL + Power BI

BE Belgium — **#1 Market**

58 orders
€23.7K revenue

DE Germany — **#2 Market**

55 orders
€22.0K revenue

ES Spain — **Hidden Gem**

48 orders
€21.6K revenue
Highest AOV: €380

FR France — **#10 by orders**

44 orders
€16.3K revenue

▶ Spain ranks 6th by orders but 3rd by revenue - Spanish customers skew toward premium products. Opportunity for targeted premium marketing.

Q3 - SELLER QUALITY & RETURN RISK

PostgreSQL + Excel

83
Total Returns

3.87
Avg Seller Rating
(Top Returners)

70%
Bronze Tier
Among Top Returners

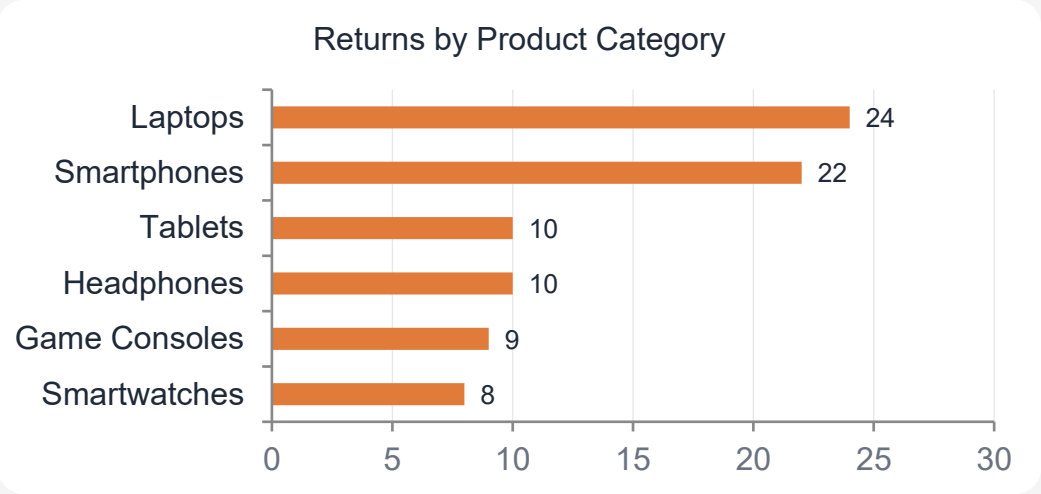
RISK SIGNALS

- 7 of top 10 returners are Bronze tier sellers
- All Bronze sellers have avg rating below 4.0
- Japan has 3 sellers in the top 10 - geographic cluster
- ChipSaver_US_15 is Silver tier but still top returner - anomaly to investigate

▶ Flag Bronze sellers with 3+ returns for quality review or listing restrictions

Q4 - RETURN RATE BY PRODUCT CATEGORY

PostgreSQL + Excel



Category	Returns	Orders	Return Rate
Laptops	24	97	**24.7%**
Headphones	10	58	**17.2%**
Tablets	10	63	15.9%
Smartphones	22	152	14.5%
Game Consoles	9	64	14.1%
Smartwatches	8	59	13.6%

CRITICAL FINDING

Laptops - The Double Risk

Laptops are the #1 revenue category (€71K) but also carry the highest return rate at **24.7%** - nearly 1 in 4 laptops is returned.

Implication:

Revenue is overstated if returns are not factored into net GMV calculations. The effective revenue contribution of Laptops may be closer to €53K after returns.

▶ Investigate Laptop return reasons - likely linked to specific Bronze sellers or Fair-grade listings

Q5 - TOP SELLER PERFORMANCE (2023-2025)

PostgreSQL + Power BI

GOLD	62 orders (51%)
SILVER	36 orders (30%)
BRONZE	24 orders (19%)

RISK FLAG

ReStore_BE_48 & EcoTech_JA_20 are Bronze sellers in the top 10 by volume - high output but elevated return risk requiring closer monitoring.

SQL IN ACTION - Q1 - REVENUE BY PRODUCT CATEGORY

back market analysis /postgres@PostgreSQL 17

No limit

Query | Query History

```sql
SELECT
    product_category,
    COUNT(order_id)                AS total_orders,
    SUM(selling_price * quantity)  AS total_revenue,
    ROUND(AVG(selling_price), 2)   AS avg_order_value
FROM orders
WHERE order_status = 'Completed'
GROUP BY product_category
ORDER BY total_revenue DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 6 | Page No:

	product_category character varying (50)	total_orders bigint	total_revenue numeric	avg_order_value numeric
1	Laptops	97	71063.46	676.31
2	Smartphones	152	63223.46	354.96
3	Tablets	63	22724.88	301.86
4	Game Consoles	64	19064.60	268.33
5	Smartwatches	59	13846.22	193.19
6	Headphones	58	10039.33	146.57

Total rows: 6 | Query complete 00:00:00.340

SQL IN ACTION - Q2 - GEOGRAPHIC PERFORMANCE

back market analysis /postgres@PostgreSQL 17

No limit

Query | Query History

```sql
SELECT
    customer_country,
    COUNT(order_id)                AS total_orders,
    SUM(selling_price * quantity)  AS total_revenue,
    ROUND(AVG(selling_price), 2)   AS avg_order_value
FROM orders
WHERE order_status = 'Completed'
GROUP BY customer_country
ORDER BY total_orders DESC;
```

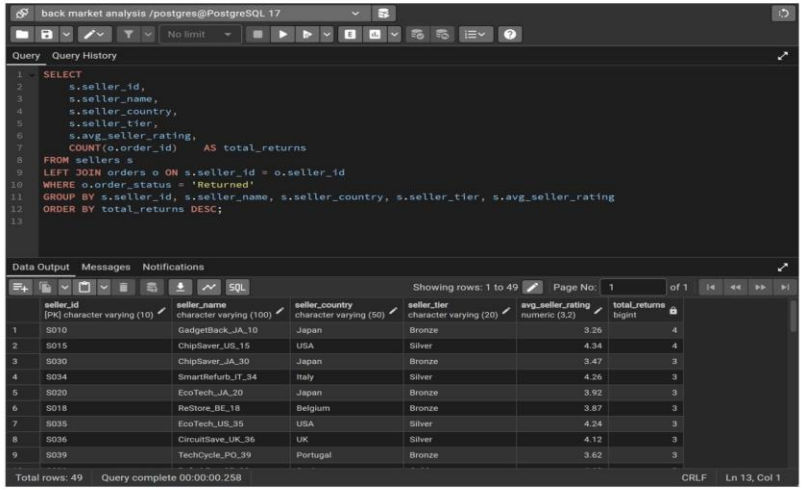
Data Output | Messages | Notifications

Showing rows: 1 to 10 | Page

	customer_country character varying (50)	total_orders bigint	total_revenue numeric	avg_order_value numeric
1	Belgium	58	23651.44	351.92
2	Germany	55	21991.65	368.50
3	Japan	53	20244.48	338.63
4	Netherlands	49	20642.49	365.28
5	UK	49	20624.18	369.01
6	Spain	48	21577.25	379.78
7	Italy	46	18874.13	376.98
8	USA	46	17842.91	367.05
9	Portugal	45	18209.66	324.79

Total rows: 10 | Query complete 00:00:00.208

SQL IN ACTION - Q3 - SELLER RETURNS



	seller_id [PK] character varying (10)	seller_name character varying (100)	seller_country character varying (50)	seller_tier character varying (20)	avg_seller_rating numeric (3,2)	total_returns bigint
1	S010	GadgetBack_JA_10	Japan	Bronze	3.26	4
2	S015	ChipSaver_US_15	USA	Silver	4.34	4
3	S030	ChipSaver_JA_30	Japan	Bronze	3.47	3
4	S034	SmartRefurb_IT_34	Italy	Silver	4.26	3
5	S020	EcoTech_JA_20	Japan	Bronze	3.92	3
6	S018	ReStore_BE_18	Belgium	Bronze	3.87	3
7	S035	EcoTech_US_35	USA	Silver	4.24	3
8	S036	CircuitSave_UK_36	UK	Silver	4.12	3
9	S039	TechCycle_PO_39	Portugal	Bronze	3.62	3

SQL IN ACTION - Q4 - CATEGORY RETURNS

back market analysis /postgres@PostgreSQL 17

No limit

Query | Query History

```sql
SELECT
    product_category,
    COUNT(order_id)     AS total_returns
FROM orders
WHERE order_status = 'Returned'
GROUP BY product_category
ORDER BY total_returns DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 6 | Page No:

	product_category character varying (50)	total_returns bigint
1	Laptops	24
2	Smartphones	22
3	Tablets	10
4	Headphones	10
5	Game Consoles	9
6	Smartwatches	8

Total rows: 6 | Query complete 00:00:00.234

SQL IN ACTION - Q5 - TOP SELLER PERFORMANCE

Query 5 of 5

```sql
SELECT
    s.seller_id,
    s.seller_name,
    s.seller_country,
    s.seller_city,
    s.seller_tier,
    COUNT(o.order_id)              AS completed_orders
FROM sellers s
INNER JOIN orders o ON s.seller_id = o.seller_id
WHERE o.order_status = 'Completed'
  AND o.order_date BETWEEN '2023-01-01' AND '2025-12-31'
GROUP BY s.seller_id, s.seller_name, s.seller_country, s.seller_city, s.seller_tier
ORDER BY completed_orders DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 60 | Page No: 1 of 1

	seller_id [PK] character varying (10)	seller_name character varying (100)	seller_country character varying (50)	seller_city character varying (50)	seller_tier character varying (20)	completed_orders bigint
1	S041	RenewTech_FR_41	France	Bordeaux	Gold	14
2	S042	BlueTech_GE_42	Germany	Frankfurt	Gold	13
3	S044	PixelRefurb_IT_44	Italy	Milan	Silver	13
4	S021	CircuitSave_FR_21	France	Lyon	Gold	13
5	S048	ReStore_BE_48	Belgium	Brussels	Bronze	12
6	S020	EcoTech_JA_20	Japan	Tokyo	Bronze	12
7	S006	CircuitSave_UK_06	UK	London	Silver	12

RECOMMENDATIONS & NEXT STEPS

01 Prioritise Laptop Category

Laptops generate the highest revenue per transaction (€676 AOV). Back Market should invest in seller recruitment, listing quality, and marketing for this category. However, the 24.7% return rate must be addressed first through grade-quality enforcement.

02 Seller Quality Intervention

Bronze tier sellers account for 70% of the highest-returning sellers. Implement mandatory quality audits for Bronze sellers with 3+ returns, consider temporary listing restrictions, and create incentive programmes to upgrade seller tier through performance improvement.

03 Target Spain for Premium Growth

Spain ranks 6th by orders but 3rd by revenue with the highest avg order value (€380). This signals a customer base that skews toward premium products. Dedicated Spanish-language marketing campaigns focused on Laptops and Smartphones could significantly increase GMV.

04 Monitor Bronze Sellers in Top 10

ReStore_BE_48 and EcoTech_JA_20 are Bronze tier sellers appearing in the top 10 by volume. High volume combined with elevated return risk creates significant GMV leakage. These sellers require enhanced monitoring and quality coaching.

TOOLS & METHODOLOGY

PostgreSQL	Built 2-table schema (sellers + orders) with 60 sellers and 600 orders across 2023-2025. Wrote 5 business queries using SELECT, COUNT, SUM, LEFT JOIN, INNER JOIN, WHERE, GROUP BY, ORDER BY, and BETWEEN to answer real marketplace questions.
Excel	Imported SQL query results into 5 structured worksheets. Applied VLOOKUP, INDEX/MATCH, SUMIF, COUNTIF, RANK, and IF formulas. Used conditional formatting - data bars, color scales, and highlight rules - to surface patterns and communicate findings visually.
Power BI	Built a 4-page interactive dashboard covering Revenue, Geography, Returns, and Seller Performance. Applied Power Query for data transformation, set correct data types, and used slicers and cross-filtering for interactivity.
PowerPoint	Translated data insights into an executive-ready presentation. Structured each slide around a business question with insight, implication, and recommendation - reflecting real analyst communication standards.

Thank You

Tushar Kumar

Business Data Analyst Portfolio - Back Market Project

- 5 SQL queries answering real marketplace business questions
- Excel analysis with formulas, conditional formatting, and cross-query insights
- 4-page Power BI dashboard with KPIs, charts, and geographic analysis
- Executive presentation structured around insight → implication → recommendation

Built with PostgreSQL | Excel | Power BI | PowerPoint